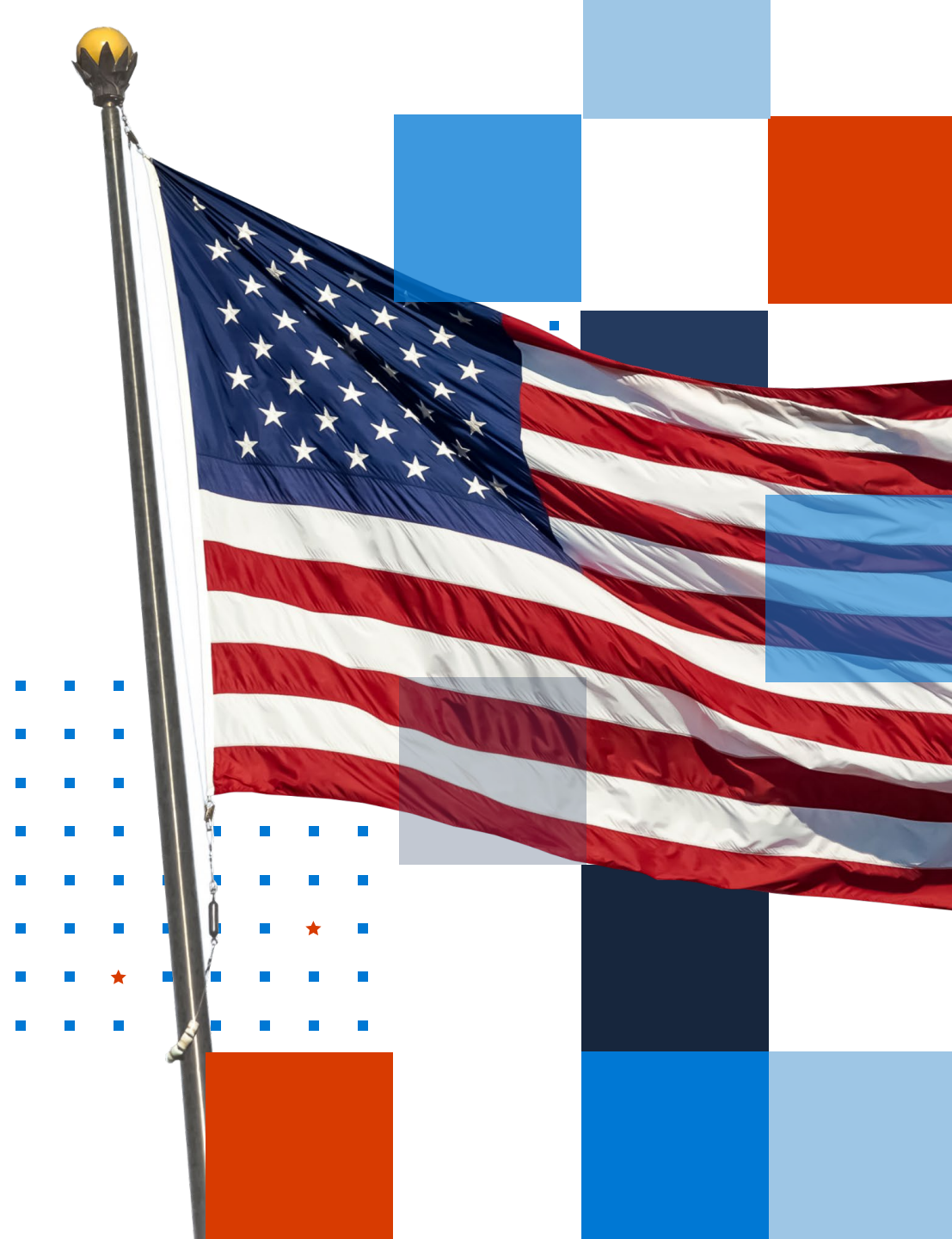
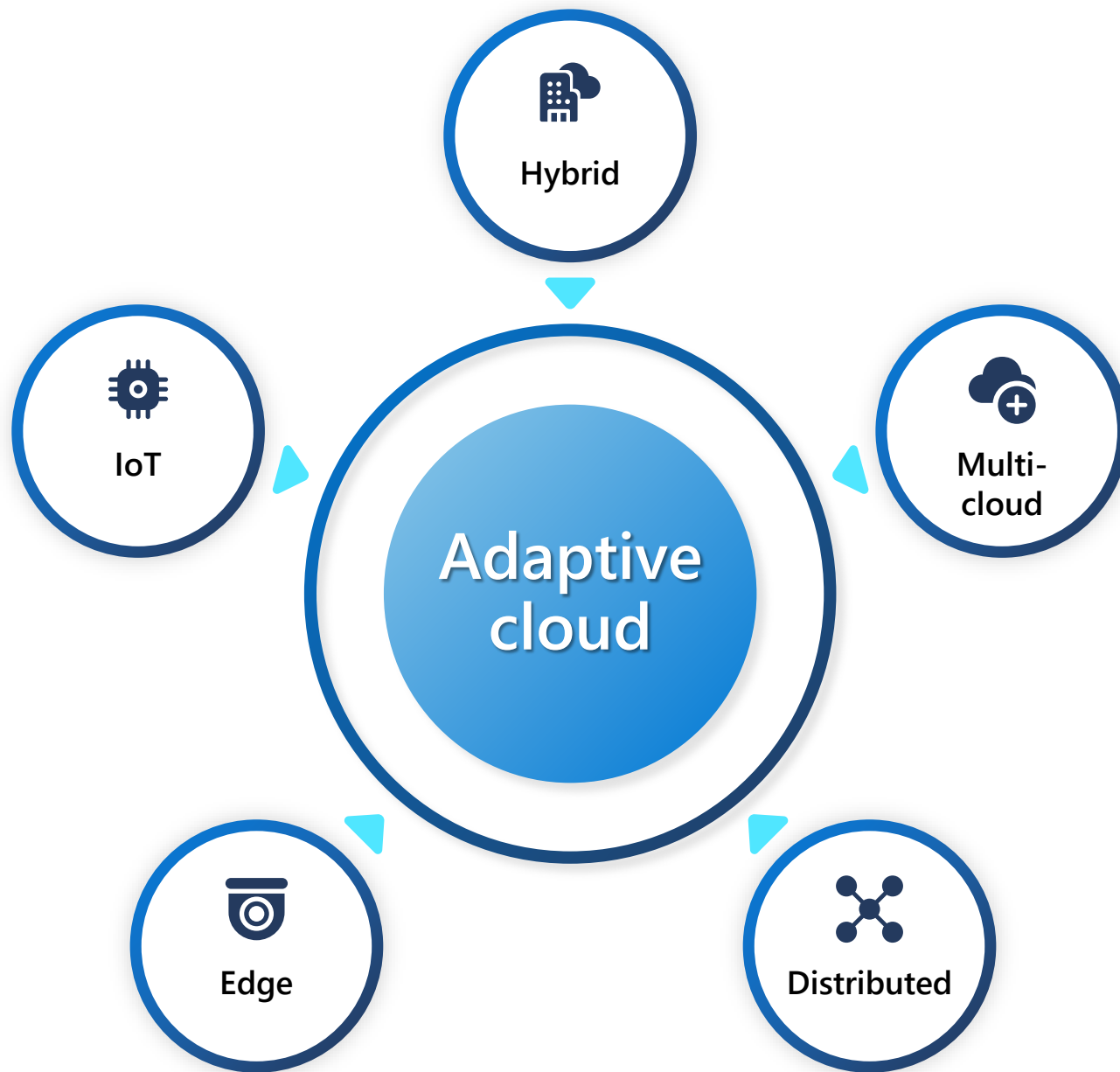


Adaptive Cloud & Extending Cloud Capabilities to the Edge

Eric Sanchez & Joe Fox





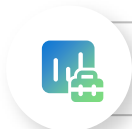
Azure's adaptive cloud approach

enabled by Azure Arc

Unify hybrid, multi-cloud, and edge infrastructure with a common operations, innovation, and insights platform.

Adaptive cloud approach enabled by Azure Arc

Cloud services and tools



Operate with AI-enhanced central **management & security**



Develop and scale **applications** across boundaries



Unify **data and AI** across a distributed estate

Global infrastructure



60+ Azure regions

Azure Local

NEW

Existing infrastructure

Operate with AI-enhanced central management & security



Configuration management and governance



Resiliency and observability



Built-in security and control



Universal AI assistant, portal, and tools

Hero services



Arc



Copilot



Policy



Defender



Monitor

Rapidly develop and scale applications across boundaries



Centralized DevOps



Simplified Kubernetes



Legacy application modernization

Hero services



Arc



Azure Kubernetes
Service



Visual Studio



GitHub

Unify data and AI across a distributed estate



Collect and process isolated data



Analyze and visualize locally and globally



Generate actionable insights with AI

Hero services



Arc



Fabric



Data services



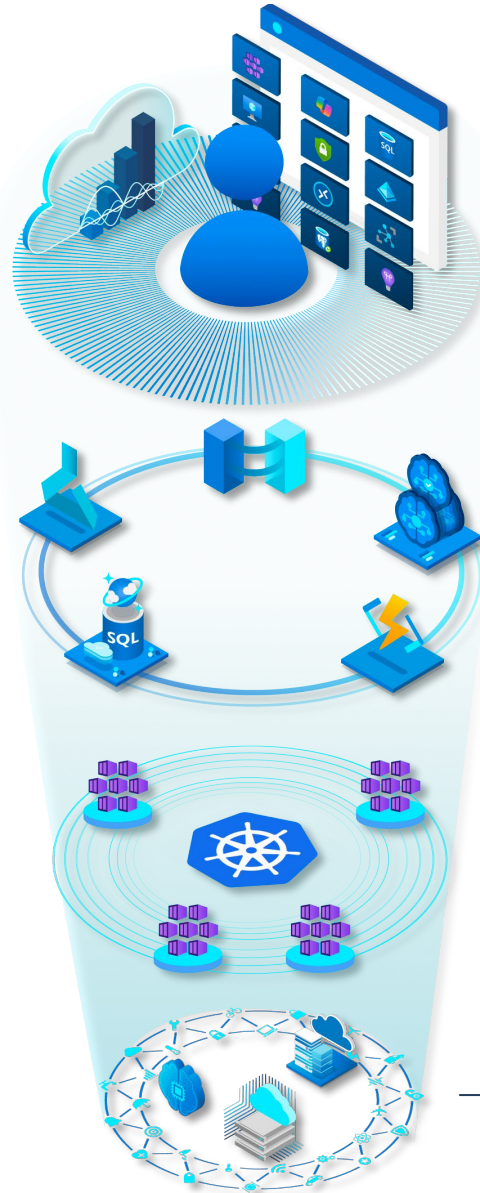
AI / ML



IoT Operations

Azure Arc

Unified management
and governance for
the adaptive cloud



Control plane

Consistent security, governance, and development practices | Git Ops | Dev Experience

Modern apps and data

Azure Arc-enabled services: Data Services | PaaS/Serverless | AI/ML

Kubernetes

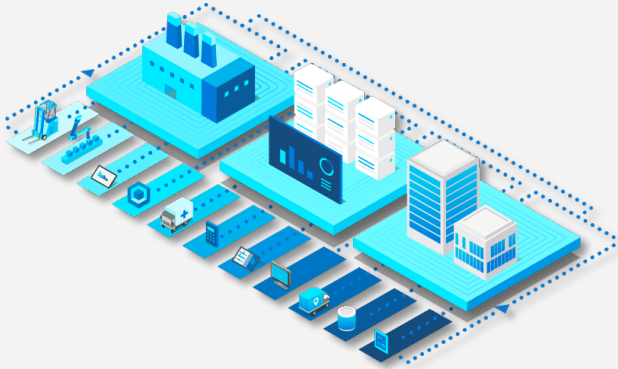
Containers | Microservices | OSS Ecosystem
AKSArc | Arc-enabled Kubernetes

Any environment

Azure Arc enabled infrastructure: Windows | Linux | Kubernetes | **Azure Local** | multi-cloud

What is driving Microsoft's Edge Strategy?

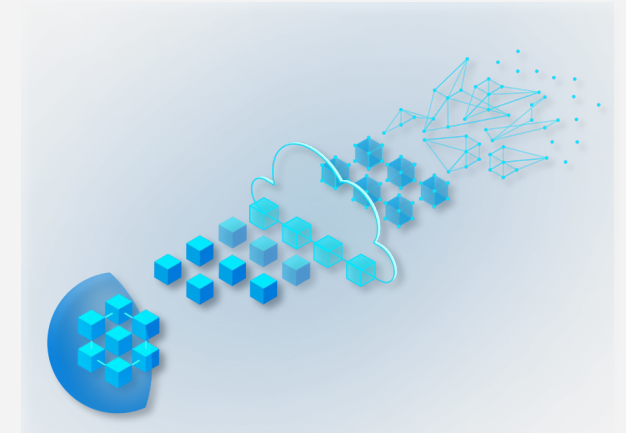
Industry innovation



Local fast/real-time processing & decision making



Innovate with AI/ML



Develop in the cloud and deploy anywhere

Just enough Azure, just where it's needed

Azure Local replaces Azure Stack HCI



Consistent software platform, Portal, and APIs



Low-spec, low-cost edge servers

Simpler, smaller hardware for light computing requirements.

NEW PREVIEW



Connected servers (formerly Azure Stack HCI)

Choose from over 100 hyperconverged server platforms from major OEMs.

✓ GA



Disconnected operations

Meet strict data residency regulations with a permanently disconnected option.

NEW PREVIEW

Existing customers of Azure Stack HCI will transition seamlessly to Azure Local with the next software update.

Azure Local

- ✓ Bare metal infrastructure solution
- ✓ Run Linux and Windows virtual machines
- ✓ Integrated optional deployment of AKS for K8s
- ✓ Built-in clustering for high availability
- ✓ Built-in software-defined storage and networking
- ✓ Scale from 1-16 nodes/cluster
- ✓ Available from 200+ validated systems from all major OEMs
- ✓ Integrates natively with Azure Arc for management



Arc-enabled
servers



Arc-enabled
Kubernetes



Arc-enabled
desktops



Arc-enabled
data services



Arc-enabled
app services

Durable scenarios for local infrastructure



Local AI
inferencing



Mission critical
business continuity



Near real-time
systems



Custom sovereignty
and regulatory
requirements

Scenarios

- Public safety
- Traffic management
- Smart city initiatives

- Transportation services
- Border patrol and national security

- Utilities and critical infrastructure
- Defense Intelligence

- Judicial systems
- Immigration services
- Revenue services

Arc-enabled Azure AI services

NEW



Local AI Search with RAG

Search and ask questions
about local data against
language models

NEW



Azure Video Indexer

Intelligent video and
audio analytics with
GenAI summarization



Azure Machine Learning *at the edge*

Train ML models in the
cloud and deploy to the
edge for inferencing

Introducing disconnected operations (preview)

NEW

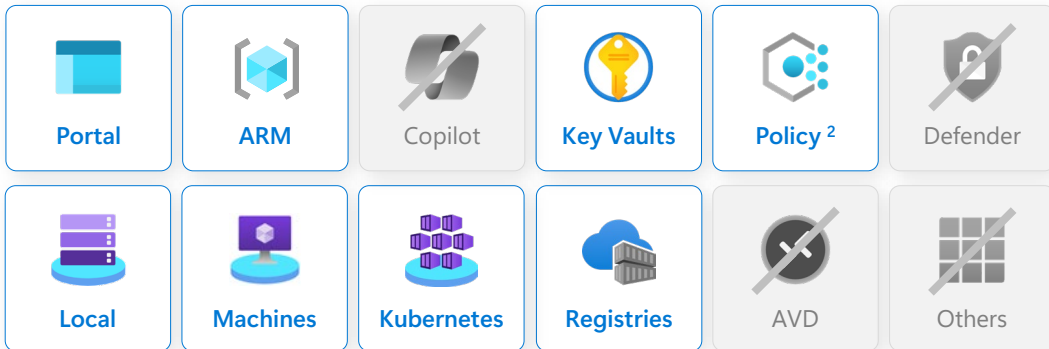


Satisfy regulatory requirements by operating permanently disconnected from the cloud



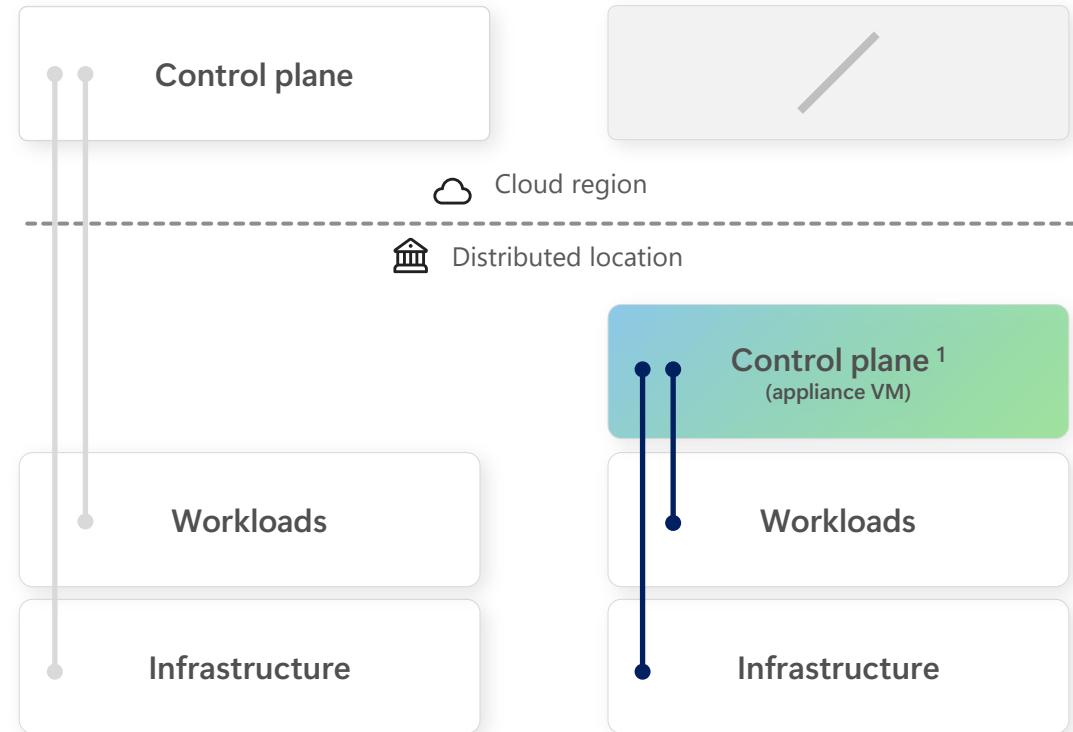
Host backend Azure resource manager, portal, and services in local appliance VM

Subset of services available:



Azure Local
(connected)

Azure Local
disconnected



1 : Available only to customers who prequalify based on industry, use case, and other considerations 2 : Partial functionality

Adaptive cloud region readiness

● Not yet planned 🔄 Preview ● GA

Adaptive cloud services	Azure Commercial	USGov	USSec	USNat
Arc enabled servers	●	●	🔄 H2 CY25	🔄 H2 CY25
Arc enabled Kubernetes	●	●	🔄 H2 CY25	🔄 H2 CY25
Azure Local 23H2	●	🔄 H1 CY25	🔄 H2 CY25	🔄 H2 CY25
Local AI search (edge RAG)	🔄 H1 CY25	🔄 H2 CY25	🔄 H1 CY26	🔄 H1 CY26
Disconnected operations	🔄 H1 CY25	🔄 H2 CY25	🔄 H1 CY26	🔄 H1 CY26
Azure IoT Operations	●	●	●	●

Thank you.

